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(1) Research Program/Project/Study Title: "Bioprospecting Boracay's Seasonal Algal Bloom; Characterization and Investigation as Possible Source of Aqua-Culture Feeds"	
Proponent Name:	Alberto Paulo I. Cercado
Academic Rank:	Assistant Professor 4
Co-Authors (with academic rank):	Stephanie Pimentel, Anacleto Cortez, Joselito Sitjar
Campus/Satellite College/Address:	Main Campus, Dayao Campus
Telephone/Cell Phone/Fax Number:	0977-282-5525
E-mail Address:	Cercadopaulo@gmail.com

Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	Application of Macroalgae as a Feed Supplement on the growth and yield of Bangus
Component Leader:	Stephanie Pimentel
Academic Rank:	FFRDC Director

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:	Malay and Dayao						
Province:	Aklan and Capiz						
Region:	6						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☒ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☒ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☐ Applied	☒ Tech. Promotion/Commercialization						
(7) Sector/Commodity Environment and Aqua Culture	(8) Discipline/Field of Specialization Environmental Engineering and Fisheries						

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(9) Rationale

This study was primarily conducted to make use of the excessive algal mass found in Boracay's shores on months before the summer where the water is calm, particularly in the months of February to April. The algal mass, consisting mainly of macroalgae species *Enteromorpha flexuosa*, *Chaetomorpha crassa* and *Ulva intestinales*, are believed to have high lipid content that makes them suitable for aqua-culture feed supplementation. In this regard, it is the intent of this study to make this waste be converted to aqua-culture feeds, specifically in the cultivation of bangus. This study can then solve two problems that will benefit primarily the Boracay Community and Fish Farmers. The first problem that will be solved should this study becomes a success would be the mitigation of the seasonal algal bloom in Boracay, and the second problem can make fish cultivation economical by the production of a cheaper pelletized feeds for aqua-culture consumption coming from raw materials that are already considered waste.

(9-A) Project Description

The over-all project strategy is to harvesting and convert the macroalgae that is seen on the shores of Boracay typically on the months of January to April into pelletized feeds that will be packaged and used as feeds in the cultivation of bangus. It will be of two phases whereby the first phase that spans 6 months will emphasize collection, drying, analysis and packaging of macroalgae to be mized later as a feed supplement in pelletized form for the cultivation of bangus. The second phase of the study will be the application and investigation of the novel pelletized bangus feed with varying macroalgae content to determine its efficiency on the yield of the bangus population. The second pahse is estimated to span for also 6 months. The first pahse will be supervised by Engr. Cercado, the progalm leader, and assisted by Engr. Cortez as the Project Staff. The second phase will be supervised by Dr. Pimentel, the project leader, and assisted by Prof. Joselito Sitjar

(10) Objectives/Statement of the Problem

Main Objective:

Harvesting and Conversion to pelletized feeds of microalgae that will be packaged and used as feeds in the cultivation of bangus.

Specific Objectives:

- 1) Collect portions of microalgae from the shores of boracay for (a) estimation of total mass and volume in the area to be specified during the course of the study, and (b) characterization for identification of different microalgae strains present and possible determination of over-all lipid composition
- 2) Collect portions of microalgae that would be dried, pelletized and packaged for use as aqua culture feeds
- 3) Collect samples from each stages of (a) processes in determination of shelf life and over-all condition of microalgae where process conditions are minimally controlled and (b) characterization for determination of over-all lipid composition
- 4) Application of the pelletized microalagae as feeds to bangus and determination of the yield of the bangus population as compared to other commercial aqua culture feeds.

(11) Review of Literature/Prior Art Search

"The green alga *Ulva lactuca* as a potential ingredient in diets for juvenile white spotted snapper *Lutjanus stellatus* Akazaki" by D. Zhu, et. Al.

This research evaluated the green alga *Ulva lactuca* as a potential ingredient in diets for juvenile white-spotted snapper *Lutjanus stellatus* Akazaki . Five test diets had similar protein and lipid levels (42 % crude protein and 8.5 % crude lipid), control diet without *U. lactuca* meal, four diets were added 5, 10, 15, and 20 % *U. lactuca* meal to investigate the effects of *U. lactuca* on growth performance, carcass composition, and enzyme activities of the fish. Each diet was assigned to triplicate groups of 30 fish for 60 days in 15 floating sea cages. At the end of feeding trail, final body weight (FBW), weight gain (WG), specific growth rate (SGR) of the fish fed 5 % UL diet were significantly higher

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than that of 10 and 20 % groups ($P < 0.05$). These results indicate that incorporation of the *U. lactuca* in the diet at 5 % for juvenile white spotted snapper does beneficial effects on the growth performance but higher level of incorporation reduce fish performance and physiological state.

(11) **Scientific Basis/Theoretical Framework**

The Theory of Waste Management represents a more in-depth account of the domain and contains conceptual analyses of waste, the activity upon waste, and a holistic view of the goals of waste management. Waste Management Theory is founded on the expectation that waste management is to prevent waste causing harm to human health and the environment. The proper definition of waste is crucial to constructing a sustainable agenda of waste management. It is largely the case that current legislation attends to existing waste. Definitions emerging from this condition may, however, conflict with the goals of waste prevention, because something that already exists cannot be prevented from arising. When material is assigned the label of 'waste', it will be treated as such; consequently, despite its explicit wish of waste prevention, implicitly, legislation essentially amasses waste. The inherent philosophical implication of such definitions is that they are not able to facilitate a sustainable waste management system. Therefore, new, dynamic definitions for waste and waste management must be sought, which can explain why waste is created and can offer an intrinsic solution for the problem.

(13) **Methodology**

Phase 1: Harvest of Macroalgae from Boracay's Seasonal Algal Bloom and Conversion to a novel pelletized meal for aqua-culture consumption.

At the start of the seasonal algal bloom in Boracay, macroalgae will be harvested using an improvised net that can scour at least 100 Liters of micro algae in 1 pull. Around 2000 liters of wet micro algae is expected to be harvested every 2 weeks during the span of the seasonal bloom. Every algae harvest, a sample will be taken for lab analysis to determine its proximate and crude content. The bulk of the algae harvest will be sun dried in Malay for 24 hours to reduce the moisture to about 20%. The dried macroalgae will be vacuum sealed in 10 kg packages and transported to the FFRDC facility of Capiz State University where it will be stored in a freezer. The freezed macrolagae will then later be used when it will be mixed with other feed components and loaded to a pelletized machine to produce the feeds for aqua-culture consumption.

Phase 2: Application of Macroalgae as a Feed Supplement on the growth and yield of Bangus

In this study, 4 treatments will be used in the application and investigation of the novel pellets as feeds of bangus. First treatment will make use of a pre identified commercial feed whom most local farmers use, the second treatment feed will make use of a novel pellet with 10 % by weight macroalgae component, third treatment will have 20% by weight macroalgae component and fourth treatment will have ovel pellet with 30% by weight macroalgae component. Over -all, there will be 12 lots of banufs population since the 4 treatment will have 3 replicates. The duration of thi study is expected to be around four months since the start of each treatment will be from bangus fingerlings until its expected optimum size that can be reched around 4 months. Computation of the yield will be done after the bangus will be harvested at the end of the fourth month.

(14) **Major Activities/Work plan** (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Activity	Inclusive Months (January - December 2023)		
Phase 1	January - June		
Phase 2		May - October	
Publication			November - December

(15) **Expected Output**

The specific Product to be produced is a novel pelletized feed for aqua culture consumption whereby the main attibue of such such ready mixed feeds is the incorporation of macroalgae that are considered ecological waste that affects tourism in Boracay Island.

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(15) Technology Transfer Activity Design

The specific Product to be produced is a novel pelletized feed for aqua culture consumption whereby the main attribute of such ready mixed feeds is the incorporation of macroalgae that are considered ecological waste that affects tourism in Boracay Island.

(16) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The study will be significant to Boracay community, the fish farmers and environmentalists. This study will help get rid of macroalgae during algal seasonal blooms in Boracay thereby improving the quality of water for leisure and improve tourism.

This study will also be beneficial to fish farmers since the study can be able to provide possibly a more complete meal and a cheaper one compared to others available in the market for their business.

This study will also provide knowledge to environmentalist and that the results of this study may be used to spring board new innovations in solving current and future environmental issues of similar nature to the study presented.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Alberto Paulo Cercado	Program Leader	Environmental Engineering
Stephanie Pimentel	Project Leader	Fisheries
Anacleto Cortez Jr	Project Staff	Mechanical Engineering
Joselito Sitjar	Project Staff	Fisheries

Percent time devoted to the Program/ Project/ Study: 30%

Program/Project Leader (in hours): 200 hrs

Program/Project Members (in hours): 200 hrs

(18) Budgetary Requirement

*****to Follow*****

(19) Literature Cited

Adhikari S, Sarkar B, Chatterjee A, Mahapatra CT, Ayyappan S (2004) Effects of cypermethrin and carbofuran on certain haematological parameters and prediction of their recovery in a freshwater teleost, *Labeo rohita* (Hamilton). *Ecotoxicol Environ Saf* 58:220–226

Barton BA, Morgan JD, Vijayan MM (2002) Physiological and condition-related indicators of environmental stress in fish. In: *Biological indicators of aquatic ecosystem stress*. American Fisheries Society, pp 111–148

Borquez A, Serrano E, Dantagnan P, Carrasco J, Hernandez A (2011) Feeding high inclusion of whole grain white lupin (*Lupinus albus*) to rainbow trout (*Oncorhynchus mykiss*): effects on growth, nutrient digestibility, liver and intestine histology and muscle fatty acid composition. *Aquacult Res* 42:1067–1078

Brinker A (2009) Improving the mechanical characteristics of faecal waste in rainbow trout: the influence of fish size and treatment with a non-starch polysaccharide (guar gum). *Aquacult Nutr* 15:229–240

Casas-Valdez M, Portillo-Clark G, Aguila-Ramírez N, RodríguezAstudillo S, Sánchez-Rodríguez I, Carrillo-Domínguez S (2006) Effect of the marine algae *Sargassum* spp. on the productive parameters and cholesterol

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content of the green shrimp, *Farfantepenaeus californiensis* (Holmes, 1900). *Rev Biol Mar Oceanogr* 41:97–105

Chandini SK, Ganesan P, Bhaskar N (2008) In vitro antioxidant activities of three selected green seaweeds of India. *Food Chem* 107:707–713

Davies SJ, Green MT, Camilleri M (1997) Preliminary assessment of the seaweed *Porphyra purpurea* in artificial diets for thick-lipped grey mullet (*Chelon labrosus*). *Aquaculture* 152:249–258

Drew MD, Borgeson TL, Thiessen DL (2007) A review of processing of feed ingredients to enhance diet digestibility in finfish. *Anim Feed Sci Technol* 138:118–136

(20) **Capsule Curriculum Vitae**

Alberto Paulo I. Cercado, PhD
Assistant Professor 4
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Roxas City, Philippines
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cercadopaulo@gmail.com

EXPERIENCE

Permanent Full-time Faculty College of Engineering and Architecture Capiz State University, Roxas City, Capiz	July 2004 - Present
Chemist Al Khaleej Sugar Refinery Jebel Ali, Dubai, UAE	Nov 2012 - May 2013
Fulltime Faculty College of Engineering/ CAS Central Philippine University, Iloilo City	June 2002 - April 2004
Test Production Supervisor Allegro Microsystems Philippines Inc. (AMPI) Bicutan, Paranaque City, Metro Manila	March 2001 - Oct 2001

EDUCATIONAL ATTAINMENT

Ph D: <u>Doctor of Philosophy in Environmental Engineering</u> Dissertation Paper “Microwave and Ultrasound-Assisted Transesterification of Microalgae Using Homogenous and Heterogeneous Alkali Catalyst” University of the Philippines - Diliman, Quezon City Recipient: USAID STRIDE for his research done at Texas A & M University (2016-2017) Recipient: ERDT DOST Scholarship (2014 -2017)	2014- 2018
Master's Degree:	2005- 2011

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Masters of Engineering in Chemical Engineering

Thesis: " *Design and Construction of a Wastewater Treatment System for the Capiz State University Main Campus (Pilot Scale)* "

Central Philippine University, Jaro, Iloilo City

Recipient: CHED Faculty Development Scholarship (2005-2007)

Tertiary Level:

1995 - 2000

Bachelor of Science in Chemical Engineering

Central Philippine University, Jaro, Iloilo City

Recipient : DOST Merit Scholarship (1995-2000)

RESEARCH PRESENTATIONS - (1997-PRESENT)

1. 30th Regional Symposium on Research and Development by WESVAARDEC (Iloilo City, Philippines; December 10-11, 2020)
Paper presented: ***Biodiesel from ultrasound assisted transesterification of WVO using sodium impregnated oyster shell as catalyst***
AWARDED AS 2ND BEST PAPER
2. 2nd International Research Conference on Science and Technology (ISATU, Iloilo City; November 27-29, 2019)
Paper presented: ***Adsorptive capacity on methylene blue of thermochemically activated biochar derived from rice husks and other agricultural wastes***
3. 1st Western Visayas Industry, Energy and Emerging Technology Research Conference by DOST R6 (Natures Village, Bacolod City, Negros Occidental; November 21-22, 2019)
Paper presented: ***Fuel Performance of Biodiesel from Microalgae***
AWARDED AS BEST PAPER
4. 1st Research, Education, Business, Science, and Arts International Conference (Davao City, Philippine; June 22-24, 2019)
Paper presented: ***Kinetics and Mechanism of Lead (II) Removal by Activated Biochar Derived from Rice Husks and Other Agricultural Wastes***
5. NOSTE International Research Conference, Training Workshop and Biennial Convention (Tacloban City, Philippines; May 17 - 19, 2019)
Paper presented: ***Biodiesel from ultrasound assisted transesterification of WVO using sodium impregnated oyster shell as catalyst***
6. 1st International Research and Education Summit (Quezon City, Philippines; Feb. 22-24, 2019)
Paper presented: ***Comparative evaluation of the Ultrasound-assisted and Microwave-assisted transesterification of WVO using sodium impregnated oyster shell as catalyst***
7. 1st International Research and Education Summit (Quezon City, Philippines; Feb. 22-24, 2019)

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Paper presented: ***Lead (II) removal from wastewater using modified biochar derived from rice husks and rice straws***

8.11th CEBU - Philippines International Conference on Environmental, Chemical, Biological and Medical Sciences (Cebu City, Philippines; January 29-30, 2019)

Paper presented: ***WVO Derived Biodiesel from Microwave Transesterification using Sodium Impregnated Oyster Shell as Catalyst***
AWARDED AS BEST PAPER

PUBLICATIONS - (1997-PRESENT)

1. Cercado, Alberto Paulo, F. C. Ballesteros, and Sergio C. Capareda. "*Biodiesel from Three Microalgae Transesterification Processes using Different Homogenous Catalysts.*" International Journal of Technology 9.4 (2018): 645-651.
2. Cercado, Alberto Paulo, Florencio Ballesteros Jr, and Sergio Capareda. "*Ultrasound assisted transesterification of microalgae using synthesized novel catalyst.*" Sustainable Environment Research 28.5 (2018): 234-239.
3. Cercado, A. P., Ballesteros, F. C., and S. C. Capareda. "*Biodiesel from microalgae using synthesized novel alkali catalyst.*" Applied Ecology and Environmental Research 15.4 (2017): 1859-1870.
4. Cercado, Alberto Paulo. "*Comparative Evaluation of the Ultrasound-Assisted and Microwave-Assisted Transesterification of WVO Using Sodium Impregnated Oyster Shell as Catalyst,*" International Research Journal of Advanced Engineering and Science, Volume 4, Issue 2, pp. 444-448, 2019.
5. Cercado, Alberto Paulo. "*Hulling Machine: Design, Fabrication and Performance*" International Journal of Information Research and Review Vol. 06, Issue, 06, pp.6371-6373, June, 2019
6. Cercado, Alberto Paulo. "*Lead (II) removal from wastewater using modified biochar derived from rice husks and rice straws ,*" International Journal of Information Research and Review Vol. 06, Issue, 05, pp.6326-6329, May, 2019
7. Cercado, Alberto Paulo. "*Design and Fabrication of Equipment for Extraction of Coconut Milk from Shells,*" International Research Journal of Advanced Engineering and Science, Volume 4, Issue 3, pp. 27-29, 2019.

PERSONAL BACKGROUND

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Home Address:	Poblacion Norte, Ivisan, Capiz, Philippines
Age:	43 years old
Birth date:	January 25, 1979
Birthplace:	Jaro, Iloilo City, Philippines
Father:	Alberto Cercado, Jr.
Mother:	Josefina Imperial Iledan Cercado
Height:	5' 5"
Weight:	140 lbs
Religion:	R. Catholic
Civil Status:	Married
Spouse:	Christine Bayhon Cercado
Child 1:	Yohan Caleb Cercado
Child 2:	Yanna Calisha Cercado